

Sales Analytics Dashboard

DATA ANALYSIS | INSIGHTS | BETTER DECISIONS



Python



SQL



Streamlit



Data Visualization



Turning
Data Into
Opportunities



Executive Summary

TURNING DATA INTO OPPORTUNITIES

This project provides a comprehensive analysis of sales performance using real business data from the Superstore dataset. Through data cleaning, exploration, and visualization, we identified key trends, profitability drivers, customer segments, and regional performance. An interactive Streamlit dashboard was developed to present the insights in a clear and actionable way, helping business stakeholders make data-driven decisions.



\$2,297,201

Total Sales



\$286,397

Total Profit



12.47%

Profit Margin



5,009

Total Orders



793

Total Customers

Key Highlights



Top Performing Category

Technology generated the highest revenue with \$836,154.



Sales Growth

Sales increased from \$484,247 in 2014 to \$733,215 in 2017.



Main Customer Segment

Consumer segment contributed the most sales with \$1,161,401.



Best Performing Region

West region achieved the highest sales with \$725,458.



Profitability Insight

Technology and Office Supplies have strong profit margins (17.4% and 17.0%).



Loss-Making Products

Top product loss: Cubify CubeX 3D Printer Double Head Print (-\$8,880).



Project Goal

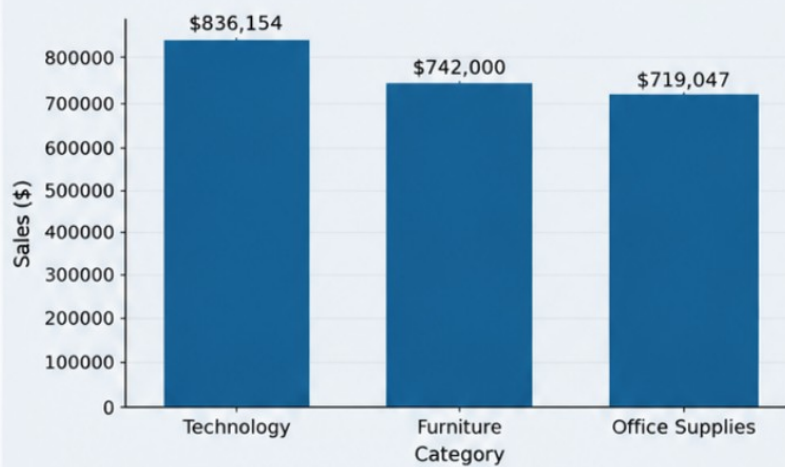
To transform raw sales data into meaningful insights and an interactive dashboard that supports strategic business decisions.

Sales Performance by Category

2014 - 2017
Superstore Dataset Analysis

Technology leads in total sales, followed by Furniture and Office Supplies, showing strong demand across all categories.

Total Sales by Category



Key Insights



Technology

generated the highest sales with \$836,154, representing the largest share of revenue.



Furniture

achieved \$742,000 in sales, showing strong demand.



Office Supplies

reached \$719,047 in sales, maintaining a stable performance.

Annual Sales Growth

Sales showed an overall increasing trend from 2015 to 2017, with the highest value in 2017.

Annual Sales Growth (2014-2017)



Yearly Sales Summary

Year	Total Sales
2014	\$484,247
2015	\$470,533
2016	\$609,206
2017	\$733,215



51.4%

Increase in sales from 2015 to 2017

Monthly Sales Trend

Monthly Performance Analysis

Sales showed fluctuations throughout the year, with the highest performance in November (\$352,461), followed by December (\$325,294).



Key Insights



Peak Month

November recorded the highest sales with \$352,461.



Strong Q4 Performance

Sales increased significantly during the last quarter.



Lowest Month

February had the lowest sales with \$59,751.

Monthly Sales Breakdown

Month	Total Sales
1	\$94,925
2	\$59,751
3	\$205,005
4	\$137,762
5	\$155,029
6	\$152,719
7	\$147,238
8	\$159,044
9	\$307,650
10	\$200,323
11	\$352,461
12	\$325,294

“

The business experiences strong seasonal demand during the final quarter of the year.



Business Takeaway

Focus on inventory planning and marketing campaigns during Q4 to maximize sales opportunities.

Profitability Analysis

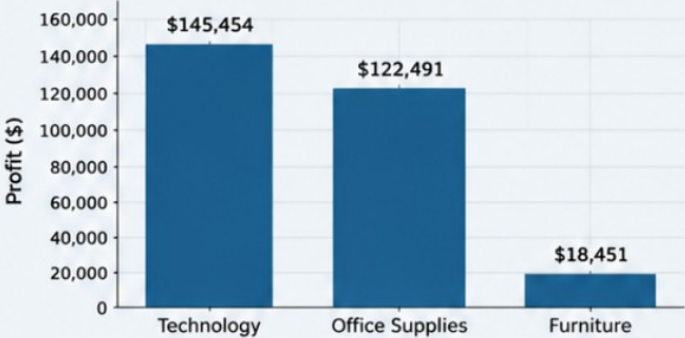
Profit by Category

Technology and Office Supplies achieve strong profitability, while Furniture shows a low profit margin despite high sales.

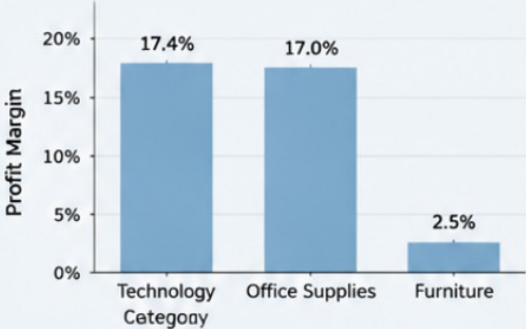


High sales do not always mean high profitability.

Total Profit by Category



Profit Margin by Category



Category Performance Summary

Category	Total Sales	Total Profit	Profit Margin
Technology	\$836,154	\$145,454	17.4%
Office Supplies	\$719,047	\$122,491	17.0%
Furniture	\$742,000	\$18,451	2.5%
Total	\$2,297,201	\$286,397	12.47%

Key Insights



Highest Margin
Technology has the highest profit margin at 17.4%.



Strong Profit Contribution
Office Supplies maintains a high profit margin of 17.0%.



Low Profitability
Furniture has a low profit margin of 2.5%, indicating potential pricing or cost issues.

Recommendations



Optimize Furniture Strategy
Review pricing, costs, and discount strategy to improve profitability.



Focus on High-Margin Categories
Increase marketing efforts for Technology and Office Supplies.



Monitor Discounts
Ensure discounts do not negatively impact profit margins.



Analyze Product-Level Profitability
Identify and address loss-making products within each category.

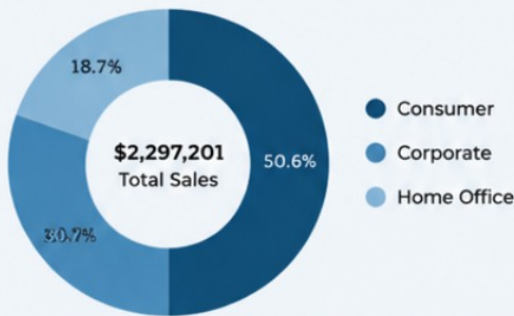


Customer & Regional Insights

Understanding Our Customers and Markets

Consumer segment contributes the highest sales, and the West region achieves the strongest performance, indicating key opportunities for targeted growth.

Sales by Customer Segment



Customer Segment Summary

Segment	Total Sales	Total Profit	Orders	Customers
Consumer	\$1,161,401	\$134,119	2,625	409
Corporate	\$706,146	\$91,979	1,752	244
Home Office	\$429,653	\$60,299	632	140
Total	\$2,297,201	\$286,397	5,009	793

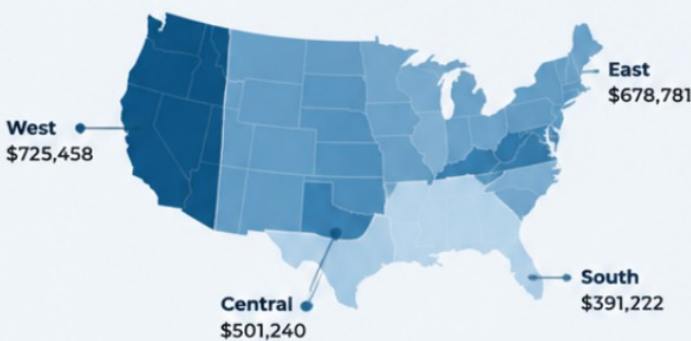
Sales by Region



Regional Performance

Region	Total Sales	Total Profit	Orders	Customers
West	\$725,458	\$108,418	1,320	245
East	\$678,781	\$91,523	1,284	217
Central	\$501,240	\$39,706	1,120	178
South	\$391,722	\$46,749	1,285	153
Total	\$2,297,201	\$286,397	5,009	793

Sales Distribution by Region



Key Insights



Consumer Segment Leads

Consumer segment contributes 50.6% of total sales with \$1,161,401.



West Region Performs Best

West region achieves the highest sales with \$725,458.



Growth Opportunities

Central and South regions show potential for growth with targeted strategies.



Focus on Customer Needs

Tailored marketing strategies can further increase customer engagement and sales.

Product Analysis

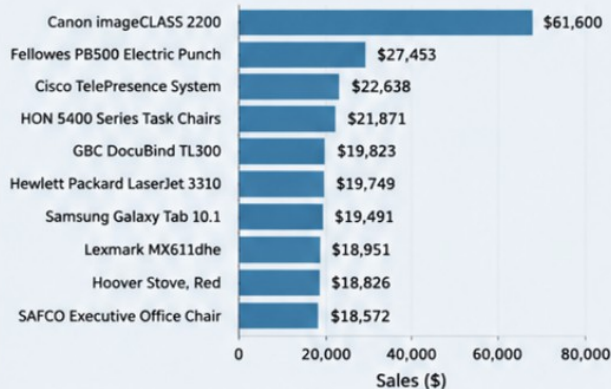
Identifying What Drives Profitability

While several products contribute significantly to revenue, some products generate losses, which affect overall profitability.

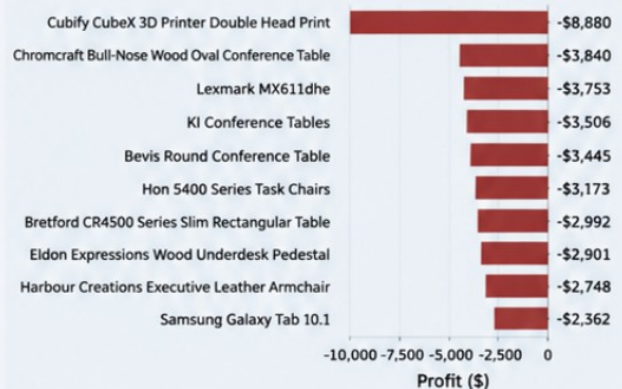
**1,859**

Unique Products

Top 10 Products by Sales



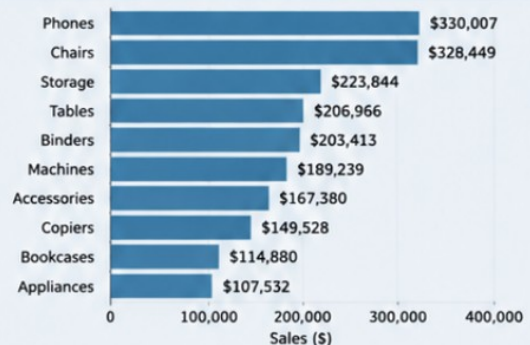
Top 10 Loss-Making Products



Worst Performing Products (by Profit)

Product Name	Category	Sales	Profit
Cubify CubeX 3D Printer Double Head Print	Technology	\$2,809	-\$8,880
Chromcraft Bull-Nose Wood Oval Conference Table	Furniture	\$1,686	-\$3,840
Lexmark MX611dhe	Technology	\$19,748	-\$3,753
KI Conference Tables	Furniture	\$1,585	-\$3,506
Bevis Round Conference Table	Furniture	\$1,763	-\$3,445

Sales by Sub-Category (Top 10)



Key Insights



Top Product

Canon imageCLASS 2200 generated the highest sales with \$61,600.



Loss-Making Product

Cubify CubeX 3D Printer Double Head Print recorded the highest loss with -\$3,880.



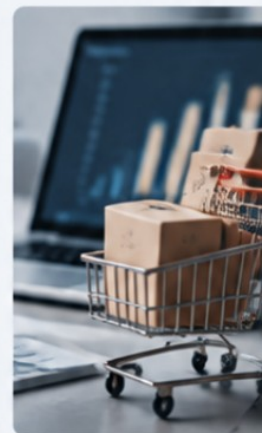
High Sales ≠ High Profit

Some high-selling products, such as Lexmark MX611dhe, still resulted in losses (-\$3,753).

Recommendation



- Review pricing and discount strategy for loss-making products.
- Focus on promoting high-performing products with strong profit margins.
- Reassess product mix in the Furniture category to improve profitability.
- Consider bundling or cost optimization for underperforming products.



Interactive Dashboard

From Data to Insights

An interactive dashboard was developed using Streamlit to explore sales performance, filter data, and gain insights easily and efficiently.



Interactive Filters

Filter by date, category, region, and segment.



Dynamic Visualizations

Explore key metrics with interactive charts.



Real-Time Insights

Get instant insights from the data.



User-Friendly Design

Clean and simple interface for easy use.



Live Dashboard

<https://sales-analytics-dashboard-2026.streamlit.app>

Explore the full interactive dashboard online.

Open Dashboard



Dataset

Superstore.csv
Orders, customers,
products, and sales data.



Tools & Technologies

Python, Pandas,
Plotly, Streamlit



Purpose

Turn raw data into
meaningful insights for
better business decisions.

Technical Implementation

Tools, Technologies and Project Structure

The project was built using Python and Streamlit, following a clean and modular structure to ensure maintainability, scalability, and easy deployment.

Technologies Used



Python
v3.10



Pandas
v2.x



Plotly
v5.x



Streamlit
v1.x



Jupyter
Notebook

Key Libraries

Library	Purpose
Pandas	Data cleaning and analysis
NumPy	Numerical computations
Plotly	Interactive visualizations
Streamlit	Web dashboard development
Matplotlib	Additional visualizations
Seaborn	Statistical visualizations

Project Structure

```
sales-analytics-dashboard/  
├── data/  
│   └── Superstore.csv  
├── notebooks/  
│   ├── 01_Data_Cleaning.ipynb  
│   └── 02_Sales_Analytics_Final.ipynb  
├── app/  
│   ├── app.py  
│   └── pages/  
│       ├── overview.py  
│       ├── sales_analysis.py  
│       ├── profit_analysis.py  
│       ├── customer_analysis.py  
│       ├── product_analysis.py  
│       └── regional_analysis.py  
├── requirements.txt  
└── README.md
```

Sample Code

```
1 import pandas as pd  
2 import plotly.express as px  
3  
4 df = pd.read_csv('data/Superstore.csv')  
5  
6 fig = px.bar(df, groupby('Category')['Sales'].sum(),  
7             title='Total Sales by Category',  
8             labels={'value': 'Sales ($)', 'Category': 'Category'})  
9  
10 st.plotly_chart(fig, use_container_width=True)
```

Development Environment



Visual Studio Code
Code editor



Jupyter Notebook
Data exploration and analysis



Git & GitHub
Version control



Streamlit Community Cloud
Deployment



GitHub Repository

View the complete source code and documentation on GitHub.

<https://github.com/your-username/sales-analytics-dashboard>



Streamlit Deployment

The dashboard is deployed using Streamlit Community Cloud.

<https://sales-analytics-dashboard-2026.streamlit.app>

Business Questions & Answers

Turning Data into Meaningful Insights

We analyzed the dataset to answer key business questions and provide actionable insights that can help improve sales performance and profitability.

Q1 Which product category generates the highest sales?



Q2 Which region performs best in terms of sales?



Q3 Which customer segment contributes the most to sales?



Q4 Which products generate the highest and lowest profits?

Top 3 Products by Profit			Top 3 Loss-Making Products		
1	Canon imageCLASS 2200	\$24,510	1	Cubify CubeX 3D Printer	-\$8,880
2	Fellowes PB500 Electric Punch	\$15,203	2	Chromcraft Bull-Nose Table	-\$3,840
3	Cisco TelePresence System	\$14,946	3	Lexmark MX611dhe	-\$3,753

While several products contribute to high profits, some products result in losses, indicating a need for pricing review or cost optimization.

Q5 How has sales performed over time?



Q6 Which sub-categories perform best?



Key Takeaways



Focus on high-performing categories and regions.



Review underperforming products and adjust strategy.



Leverage the Consumer segment for growth.



Continue monitoring trends to identify new opportunities.